

LESSON 1: THE IMPACT OF INFORMATION TECHNOLOGY ON SOCIETY

It is dangerously destabilizing to have half the world on the cutting edge of technology while the other half struggles on the bare edge of survival. —President Bill Clinton

Learning Outcomes:

At the end of this lesson, the student is expected to:

- Analyze the impact of IT on the standard of living and worker productivity.
- Identify strategies to address the negative effects of the digital divide.
- Evaluate how IT can enhance productivity by reducing costs and improving quality.
- Discuss ethical issues arising from unequal access to IT investments and their effects on productivity and quality.

TIME FRAME:

Week 12

THE IMPACT OF IT ON THE STANDARD OF LIVING AND WORKER PRODUCTIVITY

The **standard of living** differs a lot, not just between countries but also among groups within the same country. To measure this, we often use **gross domestic product (GDP) per capita**, which represents the total yearly economic output of a country divided by its population. Generally, richer, **industrialized nations** have a higher standard of living compared to **developing countries**.

In the U.S. and other developed countries, the **standard of living** has been getting better over time. However, this improvement isn't always steady because of ups and downs in the economy called **business cycles**. These cycles impact things like prices, wages, jobs, and the production of goods and services. Big disasters—like earthquakes, hurricanes, wars, or economic crises—can lower the **standard of living**.

For example, during the **Great Depression** (1929–1932), the U.S. **GDP** dropped by about 50%, and 25% of people were unemployed. In comparison, during the 2007 recession, the **GDP growth rate** fell by 6.8% at its worst point, and unemployment peaked at 10.2% in 2009.

IT Investment and Productivity

Productivity is defined as the amount of output produced per unit of input, and it is measured in many different ways. Think of it this way: in a factory, it's like measuring how many hours it takes to make one product. In a service company, it could be how much revenue one employee generates compared to their salary. Over time, countries have improved productivity not by using more resources but by working smarter—making production more efficient.

This efficiency has led to better standards of living because every hour of work creates more value. For instance, in the United States, productivity has grown by about 2% yearly over the past century. At this rate, living standards double roughly every 36 years. The Bureau of Labor Statistics monitors productivity quarterly, showing how it changes over time. For example, from 1947 to 1973, productivity rose 2.8% per year due to better management and automation. It slowed in the mid-1970s, rose again in the early 2000s, but dropped during the 2007-2012 recession.

Innovation is essential for improving productivity, and IT plays a significant role here. Companies use IT to create better products, streamline processes, and enhance services. In the 1960s, IT's impact was straightforward: automating tasks like payroll significantly reduced the need for manual labor. For example, companies with many payroll accountants could cut their workforce by implementing automated systems. The productivity gains from IT were clear and measurable.

As IT students, understanding how technology drives productivity helps you see your potential role in creating innovations that make businesses and lives more efficient.

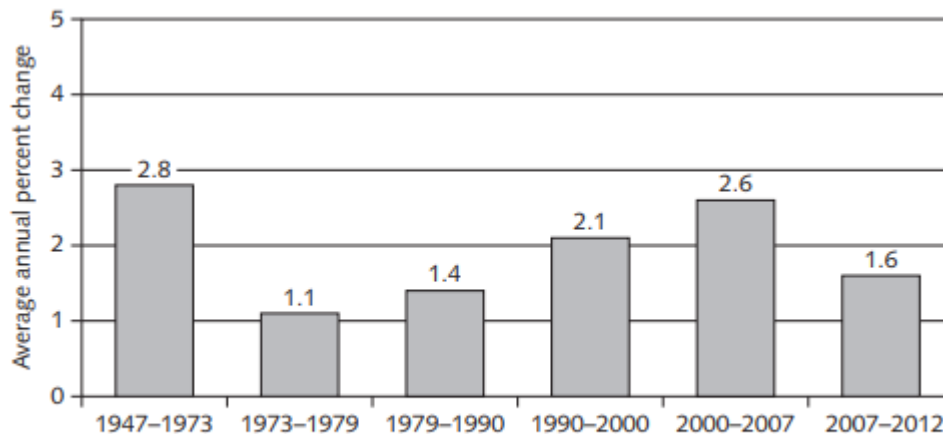


FIGURE 8-1 U.S. nonfarm labor productivity, 1947-2012

Source Line: U.S. Bureau of Labor Statistics, "Labor Productivity and Costs, 1947-2012," www.bls.gov/lpc/prodybar.htm.

Today, organizations are working to improve IT systems and business processes that have already been improved multiple times. They are also adding new IT capabilities to help workers. Many workers already use productivity tools like desktop apps, laptops, and smartphones such as BlackBerry®, Droid®, and iPhone®.

Instead of replacing workers, IT enhancements are saving small amounts of time each day. However, whether these saved minutes lead to better productivity is debatable. Some analysts believe that workers use the extra time for small tasks like answering emails they would have ignored before. These small gains make it harder to measure how much IT investments really improve productivity.

The relationship between investment in **information technology** and U.S. productivity growth is more complicated than it seems. Here are some key points:

- Between 1990 and 2000, the productivity rate increased to 2.1%, slightly above the long-term U.S. rate of 2%. However, this growth wasn't significant compared to the high rates after World War II.
- From 2000 to 2007, **labor productivity** in the U.S. increased, even though investment in IT went down. If IT investment had a direct effect on productivity, the rate should have dropped.

One explanation for this is **lag time**. When innovative IT solutions are introduced, it takes time to see significant gains. Businesses often need years to make big changes, like retraining staff, reorganizing workflows, or updating reward systems. These changes take time and can even temporarily reduce productivity. For instance, research shows that it can take 5-7 years for IT investments to significantly boost productivity.

Another explanation is that many factors beyond IT influence productivity, such as the economy, labor market flexibility, government policies, and changes in supply and demand.

IT has the potential to improve productivity, but the results are not always immediate or easy to measure. It often requires time, effort, and other changes within organizations to make a noticeable difference.

Table 8-1 summarizes fundamental ways in which companies can try directly to increase productivity.

TABLE 8-1 Fundamental drivers for productivity performance

Reduce the amount of input required to produce a given output by:	Increase the value of the output produced by a given amount of input by:
Consolidating operations to better leverage economies of scale	Selling higher-value goods
Improving performance by becoming more efficient	Selling more goods to increase capacity and use of existing resources

Source Line: Course Technology/Cengage Learning.

The following list summarizes additional factors that can affect national productivity rates:

1. **Labor productivity growth rates** change depending on whether a country is in good times (expansion) or bad times (contraction). During expansion, companies grow fast because they can produce more efficiently. But during contraction, growth slows because businesses are hesitant to invest.
2. **Outsourcing** means hiring another company to do some work for you. If the company you hire is more or less productive than your own, it can change how productive your business looks overall.
3. **Regulations** in some countries, like the United States, make it easy for companies to hire or let go of workers and start or stop businesses. This flexibility helps workers move to companies or industries where they can be more productive.
4. **Competitive markets** push businesses to innovate with new technologies to stay ahead. If a company doesn't innovate, it risks falling behind competitors.
5. In today's **service-based economy**, it's hard to measure things like the quality of customer service or consulting. For example, how do you put a number on "great service"?
6. **IT investments** don't always show immediate cost savings. Sometimes they give less obvious benefits like better quality, reliability, or customer satisfaction.

Now, about IT specifically: It's really hard to measure how much IT helps workers be more productive. Think about this: almost every business today uses IT in some form—whether it's airlines, banks, or online stores. Since no big company runs without computers, we can't compare IT-based businesses with non-IT ones anymore.

Instead of measuring IT's impact directly, businesses focus on whether their IT investments are worth it. They check if it improves their services or makes work easier. It's a bit like asking, "How much has electricity helped society?" We can't measure it exactly, but we know it's essential. The same goes for IT—it's become a basic part of how businesses operate.

Telework

Telework, also called **telecommuting**, means working away from the office. This could be at home, a client's place, a hotel, or almost anywhere. Employees use tools like texting, email, video calls, and online chats to stay connected. They access the Internet using smartphones, tablets, laptops, or similar devices to work on files, use apps, access databases, and communicate with coworkers, managers, customers, or suppliers. The main idea of telework is to stay productive no matter where you are.

Telework has become more common because of better technology, fast Internet at home, and mobile devices. Other reasons include traffic jams, high fuel costs, and concerns about pollution from cars. Plus, skilled workers often want flexible work options, like working from home sometimes.

Governments encourage telework too. For instance, Virginia met its goal of having 20% of eligible state workers teleworking by 2010. The **Telework Improvement Act of 2010** made all federal employees eligible to telework at least one day a week unless their manager decided otherwise.

Still, not all companies like telework. Some, like Yahoo! and Best Buy, cut back on it to save money, improve teamwork, and boost efficiency. Organizations need clear rules about telework—like who can do it, how tasks will be assigned, and when employees must be available. Workers also need to know what's expected of them.

TABLE 8-2 Advantages/disadvantages of teleworking for employees

Advantages	Disadvantages
People with disabilities who otherwise find public transportation and office accommodations a barrier to work may now be able to join the workforce.	Some employees are unable to be productive workers away from the office.
Teleworkers avoid long, stressful commutes and gain time for additional work or personal activities.	Teleworkers may suffer from isolation and may not really feel "part of the team."
Telework can reduce the need for employees to take time off to stay home to care for a sick family member.	Workers who are out of sight also tend to be out of mind. The contributions of teleworkers may not be fully recognized and credited.
Teleworkers have an opportunity to experience an improved work/family balance.	Teleworkers must guard against working too many hours per day because work is always there.
Telework reduces ad hoc work requests and disruptions from fellow workers.	The cost of the necessary equipment and communication services can be considerable if the organization does not cover these.

Source Line: Course Technology/Cengage Learning.

Telework isn't perfect. A report found that while most teleworkers (62%) liked it, 25% said it made no difference, and 13% felt negatively. Some teleworkers said they worked longer hours, taking time away from family. Others felt their productivity didn't really improve.

Not all jobs are good for telework. For example, jobs that need lots of face-to-face communication or direct supervision may not work well. Also, not everyone is a good fit for telework—it depends on the job and the person. **Table 8-2** and **Table 8-3** list some of the advantages and disadvantages of telework from the perspectives of employees and organizations, respectively.

TABLE 8-3 Advantages/disadvantages of teleworking for organizations

Advantages	Disadvantages
As more employees telework, there is less need for office and parking space; this can lead to lower costs.	Allowing teleworkers to access organizational data and systems from remote sites creates potential security issues.
Allowing employees to telework can improve morale and reduce turnover.	Informal, spontaneous meetings become more difficult if not impossible.
Telework allows for the continuity of business operations in the event of a local or national disaster and supports national pandemic-preparedness planning.	Managers may have a harder time monitoring the quality and quantity of the work performed by teleworkers, wondering, for instance, if they really “put in a full day.”
The opportunity to telework can be seen as an additional perk that can help in recruiting.	Increased planning is required by managers to accommodate and include teleworkers.
There may be an actual gain in worker productivity.	There are additional costs associated with providing equipment, services, and support for people who work away from the office.
Telework can decrease an organization’s carbon footprint by reducing daily commuting.	Telework increases the potential for lost or stolen equipment.

Source Line: Course Technology/Cengage Learning.

The Digital Divide

When people talk about standard of living, they are often referring to a level of material comfort measured by the goods, services, and luxuries available to a person, group, or nation—factors beyond the GDP-based measurement of standard of living. Some of these indicators include the following:

- Average number of calories consumed per person per day
- Availability of clean drinking water
- Average life expectancy
- Literacy rate
- Availability of basic freedoms
- Number of people per doctor
- Infant mortality rate
- Crime rate
- Rate of home ownership
- Availability of educational opportunities

Another indicator of standard of living is the availability of information and communications technology. The **digital divide** is a term used to describe the gulf

between those who do and those who don't have access to modern information and communications technology, such as cell phones, smartphones, personal computers, and the Internet. There are roughly 2.4 billion Internet users worldwide, but the worldwide distribution of Internet users varies greatly from region to region.

The digital divide exists from country to country (see **Table 8-4**) and even within countries—among age groups, economic classes, and people who live in cities versus those in rural areas. For example, in India, while 20 percent of urban Indians are connected to the Internet, only 3 percent of rural Indians are connected.³¹ In the United States, Hispanics and African Americans, adults living in poor households, and senior citizens are least likely to have Internet access.

TABLE 8-4 High-low Internet penetration by country within region

Region	Country with the highest Internet penetration	% of population	Country with the lowest Internet penetration	% of population
Africa	Morocco	51%	Ethiopia	1%
Americas	Falkland Islands	96%	Belize	23%
Asia	South Korea	82%	Myanmar	1%
Europe	Monaco	100%	Kosovo	20%
Middle East	Qatar	86%	Iraq	7%
Oceania/Australia	Australia	89%	Papua New Guinea	2%

Source Line: Internet World Stats, www.internetworldstats.com/stats.htm.

Many people believe that the digital divide must be bridged for a number of reasons. Clearly, health, crime, and other emergencies could be resolved more quickly if a person in trouble had easy access to a communications network. Access to IT and communications technology can also greatly enhance learning and provide a wealth of educational and economic opportunities as well as influence cultural, social, and political conditions. Much of the vital information people need to manage their career, retirement, health, and safety is increasingly provided by the Internet.

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Source Line: Internet World Stats, www.internetworldstats.com/stats.htm.

The E-Rate program discussed in the opening vignette was designed to help eliminate the digital divide within the United States. This program and others designed to increase the availability of low-cost computers, cell phones, and smartphones are discussed in the following sections.

E-Rate Program

The **Education Rate (E-Rate)** program is part of a law called the Telecommunications Act of 1996. Its full name is **The Schools and Libraries Program of the Universal Service Fund**. The purpose of E-Rate is to help schools and libraries get access to **broadband Internet services**, which are important for providing educational and informational resources.

E-Rate offers **cost discounts** of 20% to 90% for schools and libraries to pay for things like telecommunications, Internet access, and internal connections. The discount percentage depends on whether the location is urban or rural and how much financial help the students need (measured by how many students qualify for the **National School Lunch Program**).

Here's how it works: Schools and libraries apply for the discounts. Then, E-Rate reimburses the Internet service providers so that the schools and libraries can get cheaper Internet.

While E-Rate has successfully helped more schools and libraries connect to the Internet, there are some issues:

1. **Fraud** – Some people have tried to misuse the program for their own gain.
2. **Impact on learning** – A study from the University of Chicago found that although E-Rate increased Internet access for poor schools, it didn't improve student performance in subjects like math, reading, or science. This may be because schools don't know how to use the Internet effectively for learning or because the Internet isn't the best tool for improving test scores.

Despite these problems, the E-Rate program continues today with a yearly budget of \$2.3 billion because many still see its value in increasing Internet access.

Low-Cost Computers

As of June 2012, around 2.4 billion people worldwide had access to the Internet. While that seems like a big number, it means nearly **5 billion people**, or 72% of the world's population, were still **unconnected**. Most of these people come from **low-income** groups. By making **low-cost computers** available, we can help more people get online and reduce the **digital divide**—the gap between those who have access to technology and those who don't.

One Laptop per Child (OLPC)

One Laptop per Child (OLPC) is a nonprofit organization with a mission to make education more accessible for children worldwide by providing them with **low-cost laptop computers**. These laptops are specifically designed to aid in learning and are tailored to meet the needs of students in various environments, including those with limited resources.

Since its start, OLPC has partnered with **federal, state, and local governments** to distribute over **2.4 million low-cost computers** to students in **40 countries** as of 2013. Their first laptop, the **OLPC XO**, launched in 2007, featured innovative solutions for challenges in **third-world countries**. For instance, it came with a **hand crank for generating power**, which was useful in areas with unreliable electricity. This laptop was distributed at around **\$200** per unit.

The latest version, the **XO-4**, improves on these features. It is energy-efficient, requiring just **1 watt of electricity per hour**, and is sold for **\$206** with a **minimum purchase of 10,000 units**. The machine includes up to an **8 GB hard drive** and a **1.2 GHz processor**, making it capable of running essential applications. It operates on a **Linux-based operating system** and comes pre-installed with **300 learning applications** in a suite called **Sugar**. Additionally, it supports internet connectivity through **wireless or satellite communications**, ensuring access to online resources even in remote areas.

For IT students, this project showcases the power of technology in addressing global challenges. It highlights how innovation, such as low-energy consumption and alternative power solutions, can make a significant difference in improving education, especially in underserved regions.

Nepal is one of the **poorest countries** in the world, with most people earning only about **\$475 (USD) per year** and nearly **half the population (46%) unemployed**. In rural areas, many people lack access to basic services like education and healthcare.

To address this, the **OLPC program** (One Laptop Per Child) started in Nepal in **2008** as a small project in two schools. By **2011**, it expanded to **32 schools** and involved **3,300 students** from grades 2 to 6. The program's goal is to **improve education quality** and give students better access to learning materials.

To make this work, **180 teachers** were trained to use technology in their teaching and to create computer-based lessons that fit the school's curriculum. The program showed that even in rural areas, **technology-based education** can succeed when local teachers are well-prepared and trained.

ACTIVITY QUESTIONS:

- How can technology like laptops improve education in rural areas?
- What challenges do you think the OLPC program might face in a country like Nepal?

- If you were an IT professional, how would you design tools for such projects?

Classmate+

In **2006**, Intel introduced the **Classmate PC**, a low-cost laptop designed for students in **kindergarten through high school** in developing countries. The **first generation** of this laptop was priced under **\$400** and began shipping in **early 2007** to **25 countries**, including **Brazil, Chile, Nigeria, China, India, and Vietnam**.

Later, Intel collaborated with **Lenovo** to launch the **Classmate+ laptop**, which is sold in **bulk quantities** to schools and agencies in developing countries. The **Classmate models** come in two styles: the **traditional clamshell** and the **convertible tablet-style**.

It's important to note that **Intel does not build the machines** but provides the **basic design**, which is manufactured by various companies globally. Over **4 million Classmate computers** have been distributed worldwide.

Key features of the **Classmate PC** include:

- A **10-inch screen**
- Options for the **Windows 7** or **Linux operating system**
- **1 or 2 GB of RAM**
- A **built-in camera**
- **Wireless capability**
- Pre-installed **Intel Learning Series Software Suite** with educational software

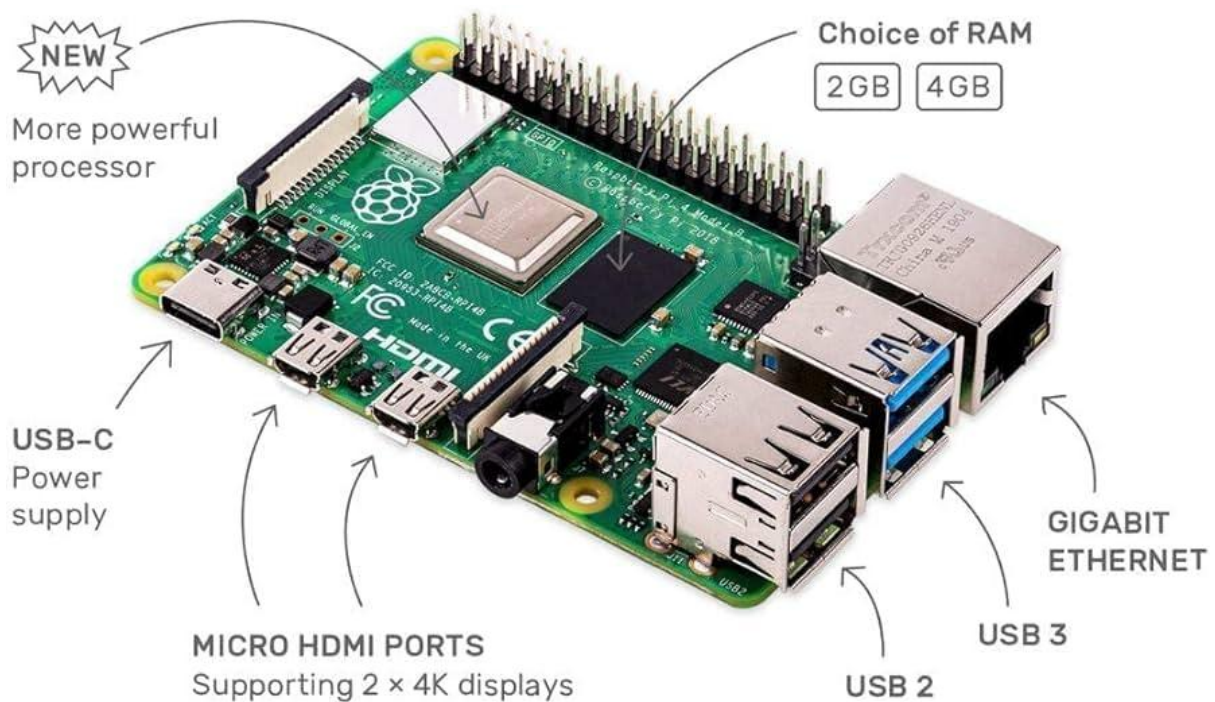
This innovation helped bridge the digital divide by making technology accessible to students in areas with limited resources. As future IT professionals, you should appreciate how such devices contribute to education and digital literacy worldwide.

Raspberry Pi

The **Raspberry Pi** is a **small** computer, about the size of a credit card, and it's also **inexpensive**, costing only \$25–\$35. It was created by the Raspberry Pi Foundation, a charity in the UK, to teach programming to young people and provide a low-cost computer for experimenting with new technology.

This computer has either **256 MB or 512 MB of RAM**, a **700 MHz processor**, **one or two USB ports**, and an **Ethernet port**. However, it doesn't include a case or a monitor. The Pi runs the **Linux operating system** and is perfect for learning coding or testing small projects.

Although its **price** is attractive, some people wonder if it's **rugged** or **portable** enough for school use or if it has all the features students might need. Also, the number of Raspberry Pi computers that will be made isn't clear yet.



Raspberry Pi

Mobile Phone: The Tool to Bridge the Digital Divide?

Many countries have struggled with the **digital divide**—the gap between people who have access to technology and those who don't. But the increasing use of **cell phones** has been a game-changer. They've become the first step in connecting more people to technology.

- **Cell phones** are much cheaper than computers, making them accessible to more people. In some cases, users can share one phone by simply swapping out a SIM card.
- They're portable and convenient—you can carry them in your pocket.
- They have a long battery life, which is crucial in areas without reliable electricity.
- Unlike computers, they're easy to use, with no steep learning curve.
- They don't need expensive apps or constant updates.
- They're reliable and require almost no technical support.

For many people, the next step is upgrading to a **smartphone**, which opens up the world of apps, the internet, and opportunities for business, education, and personal growth. Now, in almost all countries, more people own **cell phones** than computers. This widespread use has driven investments in wireless infrastructure, which in turn has encouraged organizations like banks to create apps that let users do things like store money or make transactions directly from their phones.

When **IT** becomes accessible to everyone, regardless of their background or where they live, it can create amazing opportunities:

- It helps share knowledge and culture.
- It can raise the standard of living in underdeveloped areas.

- It might even change how people see themselves in the global context, fostering better understanding and reducing issues like poverty and violence.

As IT students, think about how you can contribute to making technology more accessible and impactful. How can you design apps or systems that cater to the needs of those who only have **cell phones**? How can you innovate to make IT a force for good in underdeveloped communities? The **digital divide** is more than just a technology gap—it's an opportunity to make a difference in the world.

THE IMPACT OF INFORMATION TECHNOLOGY ON HEALTHCARE COSTS

Healthcare costs are rising fast and have become a major challenge today. Back in 2011, the U.S. spent about \$2.7 trillion on healthcare, and by 2015–2021, costs were expected to grow 6.3% yearly—way faster than inflation. One big reason is the aging population and the increasing healthcare expenses for each person.

New medical technology, like advanced diagnostic tests and treatments, plays a huge role in this. These technologies are often better but more expensive. Even if some new technologies are cheaper (like MRI machines), people use them more because they work better and are less painful (see **Figure 8.3**).



FIGURE 8-3 The development and use of new technology has contributed to increased healthcare spending

Credit: © Farferros/Shutterstock.com

Patients often overuse healthcare services because they don't directly feel the costs—insurance companies or government programs pay for most of it. This leads to

waste. Insurance companies have tried to control costs with more paperwork, but it hasn't worked well.

To fix this, we need to:

1. Help patients understand healthcare costs better.
2. Manage the costs of technology more carefully.

As IT students, you should know that better use of **IT in the healthcare industry** can reduce costs. For example, IT can streamline paperwork, track patient data efficiently, and improve communication between doctors and patients. Your skills in technology can make healthcare smarter and more affordable.

Electronic Health Records

The **healthcare industry** has been slower than other industries to use IT for better **productivity** and **efficiency**. For example, back in 2005, healthcare only spent about \$3,000 per worker on IT, compared to \$7,000 in private companies and \$15,000 in banking. But things started to change—by 2010, the healthcare industry spent over **\$88 billion** on IT. This money was used for things like **electronic health records (EHRs)**, switching to a new coding system (**ICD-10**), and creating a web portal for reporting medical device issues.

Using IT to store and manage **patient data** can make healthcare much better and faster. Think about when you visit a doctor's office and fill out the same form every time. You might wonder, "What happened to the data I gave last time?" The problem is that healthcare has traditionally relied on paper records, which are hard to organize and sometimes get lost. This causes serious issues like **diagnostic errors**, prescribing the wrong medications, or repeating unnecessary tests—all of which increase costs and risk to patients.

For instance, if someone ends up unconscious in the emergency room, doctors often have no quick way to know their **medical history**, recent treatments, medications, or **allergies**. Without this **patient data**, doctors might make risky guesses while treating them. A nationwide healthcare information network could solve this by providing instant access to vital information for any patient in any facility. Experts believe this could save lives—since mistakes in hospitals cause over 98,000 preventable deaths annually.

An **electronic health record (EHR)** is a digital version of a patient's **health-related information**. It includes everything from **demographics**, **medical history**, and **family history** to lab results, medications, and even X-ray reports. Unlike paper records, EHRs can be easily shared among **healthcare professionals**, giving a complete picture of a patient's condition and improving care.

In short, IT is transforming healthcare by replacing paper with digital systems, improving care, reducing costs, and even saving lives. As future IT experts, your skills could play a big role in developing and maintaining these life-saving systems (see **Figure 8.4**).



FIGURE 8-4 An EHR is a computer-readable record of health-related information on an individual
Credit: Courtesy of U.S. Department of Veterans Affairs.

The **Health Information Technology for Economic and Clinical Health Act (HITECH Act)** was introduced in 2009 as part of a larger economic plan to boost the U.S. economy. Its goal is to make healthcare more efficient and improve patient care using technology.

Here's what the HITECH Act does:

1. **Sets standards** for how health information is exchanged electronically across the U.S.
2. Offers **\$20 billion in incentives** to encourage doctors and hospitals to use electronic health records (EHR). This allows them to share patient information electronically, improving coordination and reducing errors.
3. Aims to **save \$10 billion** by improving care quality, reducing medical mistakes, and avoiding duplicate treatments.
4. Ensures stronger **protection of patient information**, so personal health data stays private and secure.

Doctors and hospitals can get more funding from **Medicare and Medicaid** if they show “**meaningful use**” of EHR technology. This means they must use EHRs to prescribe medicines, share patient data, and create reports on healthcare quality.

For example, a hospital with 500 beds could get up to **\$6.1 million** to adopt and maintain an EHR system. But if they didn't adopt EHRs by 2015, they risk losing **\$3.2 million** each year.

Because of the HITECH Act, many hospitals started using **basic EHR systems** with features like digital notes from doctors. Over time, more hospitals began adopting

certified EHR systems, which meet federal standards to ensure they can achieve meaningful use.

Physicians can earn up to **\$44,000 under Medicare** and **\$63,750 under Medicaid** by implementing and meaningfully using **EHR systems** (Electronic Health Records). To qualify, they must show that certified EHRs improve health outcomes, such as **e-prescribing, health information exchange**, and submitting clinical quality data. The **HITECH Act** also strengthens privacy rules, requiring patient notifications of unauthorized access, audit trails, and consent for marketing or fundraising.

Back in 2005, the **RAND Corporation** predicted EHRs would cut healthcare costs by \$81 billion annually and improve care quality, driving government investment. However, costs have risen by \$800 billion since, and there's little proof of better care or savings. Studies, like RAND's 2010 report, show limited impact, especially in large hospitals.

EHR skeptics highlight slow adoption, high costs, and usability issues. Many systems don't share data across vendors, and some think EHRs may even increase costs by simplifying billing. For example, a three-doctor practice might spend up to \$296,000 on EHRs, while large institutions like Wake Forest Baptist Medical Center reported **\$13.3 million in EHR-related expenses**. Critics question whether the investment in EHRs will truly reduce avoidable deaths or if advocates have other motives.

Use of Mobile and Wireless Technology in the Healthcare Industry

The healthcare industry, though slow to invest in IT, quickly adopted **mobile and wireless technology** due to the urgency of communications among doctors and nurses who are always on the move. For instance, doctors were early users of personal digital assistants (PDAs) for work.

Common uses of **wireless technology** in healthcare include:

- Accessing and updating **EHRs** at patients' bedsides for accurate and current data.
- Scanning bar codes on patient wristbands and medications to ensure correct drug administration.
- Communicating with healthcare staff through **wireless devices** anywhere within the facility.

This technology ensures efficiency and accuracy in patient care.

Telehealth

Telehealth uses electronic tools and telecommunications to deliver healthcare services, professional training, and administrative support remotely. Technologies like the **Internet, broadband**, laptops, **videoconferencing**, and mobile apps enhance healthcare access and monitoring. Common apps help patients manage appointments,

medications, and health data like blood pressure or weight. However, secure communication is essential to comply with regulations like **HIPAA**.

Telemedicine, a subset of telehealth, focuses on delivering medical care remotely, reducing travel and expanding access to healthcare. It has three forms:

1. **Store-and-forward:** Collects and transmits patient data (e.g., images, videos) for later evaluation by specialists. Example: A Venezuelan boy's life was saved when remote specialists identified an emergency condition.
2. **Live telemedicine:** Enables real-time interactions, such as a nurse on an oil rig consulting with emergency physicians via videoconference.
3. **Remote monitoring:** Tracks ongoing vital signs (e.g., blood pressure, glucose levels) and alerts healthcare providers to potential crises.

Telemedicine introduces legal and ethical challenges, like licensing requirements for remote care and maintaining system reliability. There's also concern about its impact on relationships between local doctors and patients.

As IT students, understanding these technologies and their secure implementation can contribute to enhancing healthcare systems.

Medical Information Web Sites for Laypeople

People, both healthy and those with illnesses, need reliable information on various medical topics to better understand healthcare services and take charge of their health. While laypeople can't become as informed as trained medical professionals, the Web provides a lot of healthcare information. Websites must ensure they publish current, reliable, and objective data.

Medical information websites, like those offering text, graphics, and images, are for informational purposes only. They are not substitutes for professional medical advice, diagnosis, or treatment. Always consult a healthcare provider with any medical questions and never ignore professional advice based on online content.

Healthcare providers, employers, and insurers also offer online tools that go beyond basic health info. These tools help compare the quality, safety, and costs of hospitals, and provide information on treatment options, risk indicators, and prices for drugs and services.

For example, if you need a hip replacement, you can find details about the surgery, available treatment options, questions to ask the physician, potential risks, and quality-of-service information about nearby hospitals.